

Kanishka Patre

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EXPERIENCE

KFC US (Yum! Brands) | *Data Engineer Co-op*

Jun 2024 - Present

- Led A/B testing and developed backend components for a sales alert system using SQL and Python to identify hourly store-level anomalies; integrated data analysis with Databricks alerts, reducing incident response time by 20%.
- Created sentiment analysis pipeline using LLaMA 3.1-70B and artificial intelligence tools for prompt engineering to process 100+ customer reviews across 20+ business domains, delivering structured insights and actionable KPIs that guided product and service improvements
- Implemented CI/CD pipelines for Power BI using GitLab, Microsoft Fabric APIs, and Python, reducing deployment time from 30+ minutes to under 5 minutes (83% improvement) and eliminating connection string errors across environments.
- Directed the migration of 73+ tables from Hive Metastore to Unity Catalog and Snowflake using parallel processing, dbt, and Data Modeling principles; ran data integrity checks to validate migrations, cutting data duplication and reducing storage costs by 50%.
- Optimized ETL (Extract, Transform, and Load) pipelines on Databricks and Spark, integrating Delta Lake, Z-ordering, and streamlined time logic to improve data refresh speed by 35%, enhancing data scalability and reliability for executive dashboards.

Computational Lab, UMass Dartmouth | *AI Engineer - Apprenticeship*

Oct 2023 - May 2025

- Engineered a multimodal deep learning pipeline for breast cancer detection, engineering patient-level data stratification to prevent leakage across 5,000+ images, ensuring reproducibility for downstream deployment
- Architected PyTorch Lightning training infrastructure with automated checkpointing, GPU acceleration, and TensorBoard experiment tracking, cutting training time 60% and enabling faster iteration cycles
- Instrumented model monitoring and logging aligned with IT general controls, reducing error rates 15% and supporting production-grade reliability standards

Rakuten | *Analytics Intern*

Feb 2022 - May 2023

- Programmed secure, RESTful APIs for data authentication and access control, incorporating IT general controls to increase data retrieval speed by 20% and reduce unauthorized access incidents by 30%
- Analyzed SQL and Python scripts to streamline data workflows, decreasing data processing time by 22% and facilitating real-time data analytics for operational improvements to the data warehouse table

Samsung R&D Institute | *Machine Learning and Data Intern*

Aug 2019 - Jan 2020

- Constructed a scalable Spark pipeline to process 8.7K+ audio files for speech detection with 92% accuracy
- Collaborated with ML engineers to refine feature extraction and model deployment, increasing accuracy by 5% and supporting operational analytics, business intelligence and product decisions

SKILLS

Programming and Database: Python, SQL, JavaScript, Java, C++, HTML, R, MATLAB, MongoDB, Snowflake

Frameworks and Libraries: A/B Testing, Regex, NodeJS, ReactJS, Flask, Seaborn, Plotly, Spark, NLTK, NumPy, Pandas, CoreML, Scikit-learn, PyTorch, Keras, TensorFlow

Platform Tools: Kubernetes, GitHub Actions, Git, Postman, Jira, CircleCI, Jenkins, Jupyter, Airflow, Docker, R Shiny, Power BI, Tableau, Azure ML, Microsoft Azure, Google Cloud Platform (GCP), Amazon Web Services (AWS), GitLab, Databricks, QlikView, Linux

Machine Learning Systems: Deep Learning (CNN, RNN, LSTM), Boosting, Bagging, Clustering, Classification (Logistic, XGBoost, Decision Tree, Random Forest, SVM), Regression (Logistic, Polynomial, Ridge/Lasso)

Agentic AI/LLM: Artificial Intelligence Tools, AutoGen, multi-agent orchestration, OpenAI APIs, LangChain, LLM, Claude Code

EDUCATION

University of Massachusetts | *Master of Science, Data Science*

Dartmouth, MA

BMS College of Engineering | *Bachelor of Engineering, Computer Science*

PROJECTS

AI-Assisted Legal Document Analysis for Professional Workflows

- Built an NLP system using Python, Transformers, and Azure ML to process 50K+ documents, integrating Power BI dashboards to extract insights
- Incorporated predictive modeling and data integrity analysis to highlight high-priority documents, using Claude Code to accelerate development, reducing document review time by 30%

Accessible Information Technology: Automated Audio Content Generation

- Designed a data pipeline using Python and Snowflake with IT general controls for data quality and integrity in ETL of 10K+ Reddit and Twitter content records
- Documented technology processes and controls while following data warehousing best practices to ensure reliable storage, improving processing speed by 25%